

TIC 158864044

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.715552 d
Epoch (t0) 1600.2586 BTJD
Duration 0.1700 d
Depth 16.358 % (163,581 ppm)
SNR 187.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

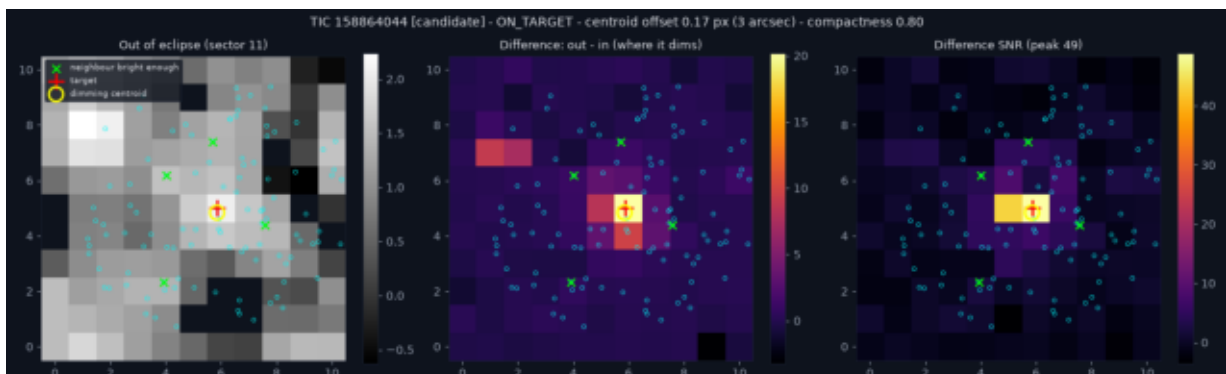
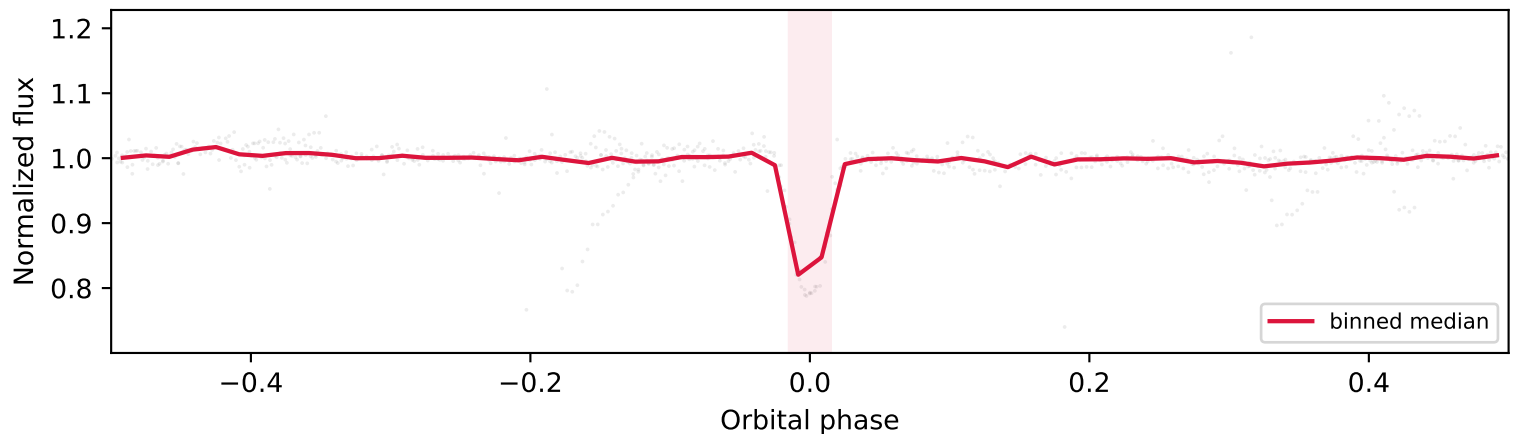
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.10 (gate ≥ 0.50)
Robust depth sigma 69.7 (gate ≥ 7.0)
Odd/even z-score 1.38 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 80.6 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 48.88 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.17 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 158864034
1.79 px away, dTmag +1.12 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.